



Negative attitudes about medications for opioid use disorder among criminal legal staff

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ARTICLE INFO

Keywords:

Medication for opioid use disorder
Criminal legal system
Staff
Stigma
Addiction
Attitudes

ABSTRACT

Background: Stigma is a barrier to the treatment of opioid use disorder (OUD) in the criminal legal system. Staff sometimes have negative attitudes about medications for OUD (i.e., MOUD), but there is little research on what drives these attitudes. How staff think about criminal involvement and addiction may explain their attitudes toward MOUD.

Methods: A convenience sample of U.S. criminal legal staff (e.g., correctional/probation officers, nurses, psychologists, court personnel) were recruited via online methods ($N = 152$). Participants completed an online survey of their attitudes about justice-involved people and addiction, and these were entered as predictors of an adapted version of the Opinions about Medication Assisted Treatment survey (OAMAT) in a linear regression, controlling for sociodemographics (cross-sectional design).

Results: At the bivariate level, measures capturing more stigmatizing attitudes toward justice-involved people, believing addiction represents a moral weakness, and believing people with addiction are responsible for their actions and their recovery were related to more negative attitudes about MOUD, whereas higher educational attainment and believing addiction has a genetic basis were related to more positive attitudes about MOUD. In a linear regression, only stigma toward justice-involved people significantly predicted negative attitudes about MOUD ($B = -.27, p = .010$).

Conclusion: Criminal legal staff's stigmatizing attitudes about justice-involved people, such as believing they are untrustworthy and cannot be rehabilitated, contributed significantly to negative attitudes about MOUD, above their beliefs about addiction. The stigma tied to criminal involvement needs to be addressed in attempts to increase MOUD adoption in the criminal legal system.

1. Introduction

The U.S. is experiencing an intersecting incarceration crisis and opioid epidemic (NIDA, 2021a). Although rates of incarceration are declining, change has been slow. There are still two million people incarcerated in U.S. jails and prisons and nearly five million individuals supervised on probation and parole at any given time (Maruschak & Minton, 2020). Using opioids increases risk for criminal involvement (Winkelman et al., 2018), and approximately 26% of people with heroin use disorder are incarcerated at some point (Boutwell et al., 2007). Moreover, those released from prison are more likely to die from an opioid-related overdose compared to the general population, particularly within the first two weeks after release (Binswanger et al., 2013). Effectively treating people with opioid use disorder (OUD) while in the criminal legal system is a high priority.

Medications for opioid use disorder (MOUD), including methadone, buprenorphine, and naltrexone, are highly effective for treating OUD. Methadone and buprenorphine are full and partial opioid agonist medications respectively, meaning they activate opioid receptors in the brain similar to other opioids. Extensive research shows these medications increase treatment engagement and reduce craving, withdrawal symptoms, and illicit opioid use, thereby reducing risk of overdose and mortality by 40–50% (Mattick et al., 2009; Timko et al., 2016; Larochelle et al., 2018; Santo et al., 2021; Wakeman et al., 2020). In contrast, long-acting injectable naltrexone is an opioid antagonist, which binds to receptors and prevents the effect of other opioids (NIDA, 2021b). Naltrexone is efficacious for treating OUD, although low rates of retention may limit its ability to prevent overdose and mortality (Tanum et al., 2017; Jarvis et al., 2018; Lee et al., 2018; Nunes et al., 2021). Although national guidelines state that MOUD should be available to justice-involved people (Blanco & Volkow, 2019; SAMHSA, 2019), and research documents numerous benefits of MOUD for justice-involved people (Jarvis et al., 2018; Macmadu et al., 2020; Sugarman et al., 2020; Moore et al., 2019), MOUD is rarely offered in criminal legal settings (Friedmann et al., 2012; Grella et al., 2020;

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Rich et al., 2005). For example, only 3.6% of incarcerated people with OUD received buprenorphine in 2021 (Thakrar et al., 2021).

1.1. Attitudes toward MOUD in the criminal legal system

Negative attitudes about MOUD are common in the criminal legal system, impacting whether MOUD is offered (Andraka-Christou et al., 2019; Friedmann et al., 2012; Rich et al., 2005; Richard et al., 2020; Sharma et al., 2017), as well as how people who engage with MOUD are viewed and treated (Cioe et al., 2020; Finlay et al., 2020; Polonsky et al., 2016). Criminal legal staff often believe MOUD is “trading one drug for another” (Streisel, 2018) and “rewards criminals” for being drug users (Andraka-Christou et al., 2019). Many criminal legal staff also report concerns about diversion and misuse of MOUD (Friedmann et al., 2012; Matusow et al., 2013; Mitchell et al., 2016; Nunn et al., 2009; Krawczyk et al., 2017; Richard et al., 2020). While both community (e.g., Cioe et al., 2020; Richard et al., 2020) and criminal legal staff (Andraka-Christou et al., 2019; Andraka-Christou & Atkins, 2020; Richard et al., 2020) express negative attitudes towards MOUD, the latter express more support for the punishment of drug use, know less about MOUD efficacy, and their beliefs align more with abstinence-only approaches (Gjersing et al. 2007; Friedmann et al., 2012; Friedmann et al., 2015; Matusow et al., 2013). Criminal legal staff have reported a preference for psychosocial treatment for OUD (Winetsky et al., 2020) and drug-free detoxification (Nunn et al., 2009; Rich et al., 2005), compared to MOUD, and are less likely to refer their clients to MOUD than community providers (Krawczyk et al., 2017; Streisel, 2018). These types of attitudes towards MOUD prevent its uptake in legal settings (Grella et al., 2020) and have been implicated in the high rates of death among legally-involved people with OUD (Wakeman & Rich, 2018). It is essential to understand what drives attitudes about MOUD among criminal legal staff.

1.2. What contributes to attitudes toward MOUD?

Research identifying what predicts attitudes about MOUD is mostly limited to medical providers, showing that decreased knowledge of MOUD efficacy, less training/experience with treating OUD, and believing in abstinence-based versus harm-reduction approaches predict negative attitudes toward MOUD (Cioe et al., 2020; Madden, 2019; Olsen & Sharfstein, 2014; Shidlansik et al., 2017). In contrast, medical providers who have more formal education endorse more positive attitudes toward MOUD (Shidlansik et al., 2017).

Among criminal legal staff, prison medical directors and probation/parole officers' attitudes towards MOUD have been found to be more positive if they supervised clients who were successful on MOUD (Mitchell et al., 2016; Rich et al., 2005). Working in a rural area and having less MOUD education has been associated with more negative attitudes, and attitudes are more negative among law enforcement officers compared to other court personnel (Andraka-Christou et al., 2019). Also, less training and education (Streisel, 2018), disbelief in the efficacy of MOUDs (Friedmann et al., 2012; Friedmann et al., 2015; Matusow et al., 2013), concerns around diversion/misuse (Matusow et al., 2013), and perceived structural barriers (e.g., security concerns, state/local regulations, cost; Friedmann et al., 2012; Matusow et al., 2013; Nunn et al., 2009) predict less intent to refer clients to MOUD among legal staff.

1.2.1. Stigma-related attitudes as predictors

Stigmatizing attitudes about addiction and criminal involvement are common across systems and settings (Belenko et al., 2018; Van Boekel et al., 2013; Barnett et al., 2018; Rade et al., 2016; Hirschfield & Pi-quero, 2010; Kjelsburg et al., 2007) and may drive attitudes toward MOUD. For example, among stakeholders in medical and legal settings, beliefs that people choose to be addicted relate to negative attitudes about MOUD (Richard et al., 2020). In addition, Madden (2019) found

through qualitative interviews with addiction treatment professionals that stigma towards MOUD overlaps with stigma towards people that misuse opioids, and other studies describe MOUD and OUD stigma as intersecting (Smith et al., 2020). Although research has yet to examine the impact of stigma toward criminal involvement on MOUD attitudes, studies show that criminal legal staff who have more negative attitudes about justice-involved people are more likely to support punishment over treatment (Schaefer & Williams, 2018). Little research is being done to parse out the relationship between stigma broadly (e.g., attitudes about substance use, criminal involvement) and attitudes towards interventions (e.g., MOUDs) among criminal legal staff.

1.3. Present study

There are multiple types of stigmatizing attitudes, including stigma toward individuals (e.g., people with opioid use disorder, those involved in the legal system) and stigma surrounding treatment (e.g., MOUD), but research has yet to examine how different types of stigmatizing attitudes may impact attitudes toward MOUD among criminal legal staff. The current study used a cross-sectional design to examine the extent to which attitudes towards criminal involvement and addiction are drivers of attitudes towards MOUD among criminal legal staff. We hypothesized that more negative attitudes about criminal involvement, as well as beliefs that addiction is controllable, a weakness, and that people are responsible for recovery, will be related to more negative attitudes about MOUD.

2. Materials and methods

2.1. Participants and procedures

Participants were staff working in various U.S. criminal legal settings ($n = 152$) recruited by advertising the study with personal contacts as well as on online platforms, including Facebook groups, subreddits, and listservs (e.g., American Probation and Parole Association, American Psychological Association) from June to August 2020. Eligible participants lived in the U.S., were over age 18, and self-identified as working at a “criminal justice agency,” which was described as “correctional and probation officers, judges, counselors, psychologists, psychiatrists, nurses/medical staff, or administrative staff who are employed in jails, prisons, probation sites, or courts.” Participants consented and completed measures as part of a larger online survey that took approximately 45–60 min (see study measures in Appendix). Four attention check questions were used to gauge valid responding, requiring participants to enter a specific response (e.g., “For this question, please mark strongly agree.”). At the end of the survey, participants provided their email address and were compensated with a \$15 e-gift card. All data were examined for validity, with those representing odd/unusual patterns of responses (e.g., responding all “1s” across measures) or failing more than one attention check excluded. Participants without contact with justice-involved people at their job were also excluded from analyses. All procedures received approval by the university's institutional review board.

2.2. Measures

2.2.1. Sociodemographic characteristics

Participants provided demographic information, including age (continuous), race/ethnicity (i.e., Caucasian/White, African American/Black, Hispanic/Latino, American Indian/Alaska Native, Asian/Pacific Islander, other), educational attainment (i.e., less than high school diploma, high school diploma/GED, vocational school/degree, some college, bachelor's degree, master's degree, doctoral degree), as well as characteristics about their work setting, including state of employment, type of justice agency (i.e.,

jail/detention center, state or federal prison, probation/parole agency, drug recovery court, other), position title (i.e., correctional officer, counselor/therapist/psychologist, psychiatrist, social worker/case manager, nurse, probation officer, administrative staff, other), years employed with their agency (continuous), number of years worked with justice-involved individuals (continuous), and amount of direct contact with justice-involved individuals at their job (i.e., every day, every other day, a couple times per week, no direct contact).

2.2.2. Stigma toward criminal involvement

The Attitudes Toward Prisoners scale (ATP), developed by Melvin et al. (1985), assessed how staff perceive justice-involved individuals. This scale includes 36-items composed of 19 negatively worded statements that capture common stereotypes about people with criminal involvement (e.g., “Prisoners never change” and “Most prisoners are stupid”) and 17 positively worded statements (e.g., “Prisoners will listen to reason” and “Prisoners have feelings like the rest of us”). Responses were rated on a 5-point Likert scale, ranging from 1 (disagree strongly) to 5 (agree strongly). After reverse coding positively worded statements, responses were summed to create a composite score ranging from 36 to 144, with higher scores indicating more stigmatizing attitudes. The ATP has been widely used as a valid and reliable way to assess stigmatizing attitudes about justice-involved people (Kjelsberg et al., 2007; Ware et al., 2012) and also displayed excellent internal consistency in this sample ($\alpha = .96$).

2.2.3. Beliefs about addiction

The Addiction Belief Inventory (ABI; Luke et al., 2002) is a 30-item measure that was used to assess participants' beliefs about people with substance use problems. The ABI was developed by merging frequently used models of addiction (e.g., the Disease Model, 12-step models, etc.) to assess a broad range of addiction beliefs (Luke et al., 2002). The ABI consists of eight subscales capturing beliefs that people can control their substance use (*inability to control* $\alpha = .70$), that addiction is a disease and the only way to control it is by abstinence (*chronic disease* $\alpha = .22$), that people can only decrease their substance use with professional help (*reliance on experts* $\alpha = .79$), that people are responsible for their substance use (*responsibility for actions* $\alpha = .70$), that people are responsible for their own recovery (*responsibility for recovery* $\alpha = .73$), that people are predisposed to addiction due to genetics (*genetic basis* $\alpha = .70$), that people cope with stress by using substances (*coping* $\alpha = .90$), and that people who use substances are morally weak (*moral weakness* $\alpha = .81$; Luke et al. 2002). Participants rated their responses on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). Items on the inability to control and responsibility for actions scales were reverse coded. Each subscale was averaged with higher scores indicating more agreement with the subscale. The ABI is most often analyzed by subscale rather than a total score (Luke et al., 2002), and studies have found acceptable reliability among most subscales ranging from $\alpha = .61$ to $.83$ (Jordan et al., 2002; Luke et al., 2002). Due to low reliability in previous studies and our sample, the chronic disease subscale was removed from the current analyses.

2.2.4. Attitudes toward medications for OUD

An 11-item adapted version of the Opinions about Medication-Assisted Treatment scale (aOAMAT; Friedmann et al. 2015) evaluated staff's perceptions about the use of methadone and buprenorphine among justice-involved individuals. Responses were rated on a Likert scale where 1 = strongly disagree and 5 = strongly agree, and we added a sixth response option if they had never heard of the medications. Participants who endorsed that they had never heard of methadone and/or buprenorphine were excluded ($n = 11$). Like the original scale, methadone and buprenorphine were included in the items' wording; however, rather than separately asking about each medication, both were included in the same item to reduce assessment burden (e.g., “Methadone and/or buprenorphine should be available as a lifelong

treatment option”). This collapsed the 18 original items into 9. Two additional items regarding access to medication during incarceration were added (i.e., “Prisoners do not need methadone or buprenorphine during incarceration because they cannot access drugs during incarceration” and “Prisoners do not need methadone or buprenorphine services after they get released because they have not used drugs while they were incarcerated”). After negatively worded items were reverse coded, all items were averaged creating a composite score ranging from 1 to 5, with higher scores indicating more favorable perceptions. This scale had good internal consistency in this sample ($\alpha = .82$).

2.3. Data analysis plan

SPSS Version 26 was used to analyze the data, with missing data being handled using listwise deletion. Bivariate correlations were examined for all study variables. Consistent with prior research, 7 ABI subscales (chronic disease subscale was excluded for low internal consistency) were considered for analysis rather than the total score (Luke et al., 2002). A multiple linear regression was conducted examining predictors of aOAMAT scores, including the total ATP and 7 ABI subscales, as well as any demographic or work characteristic factors that were significantly associated with attitudes toward MOUD at the bivariate level. Statistical significance was determined by $p < .05$ and interpreted in conjunction with effect sizes.

3. Results

Two hundred seventeen participants initiated the survey; 154 of these people provided complete data and answered at least 3 out of 4 attention checks correctly; 52 people consented but provided insufficient data to analyze and were excluded, and 11 people were removed due to failing more than one attention check. An additional 2 participants indicated that they had no contact with justice-involved individuals as part of their work and were excluded. The final sample contained 152 criminal legal staff (Table 1). The sample was predominately White (80.9%) and ranged in age from 22 to 55 years ($M = 37.5$, $SD = 8.1$). The majority of the sample worked in state or federal prisons (84.2%) and had daily contact with justice-involved individuals at work (92.1%). Staff reported working in their current agency from 6 months to 31 years ($M = 104$ months, $SD = 87.1$) and working with justice-involved individuals from 9 months to 31 years ($M = 132.7$ months, $SD = 87.2$). For education, 3.9% had a high school diploma, 2.6% had a vocational degree, 11.2% had some college but no degree, 24.3% had a bachelor's degree, 23.0% had a master's degree, and 34.9% had a doctoral degree. Regarding position types, 13.2% were correctional officers, 17.1% were bachelor- or masters-level counselors or therapists, 36.2% were psychologists, 6.6% were social workers or case managers, 6.6% were health care workers, 3.3% were probation/parole officers, 13.8% were administrative staff, 2.0% were educators, and 1.3% held other staff positions. A total of 11 people were missing data on either the ATP, aOAMAT, or ABI subscales, resulting in a final analyzed sample of 141 participants.

On average, criminal legal staff scored 3.41 on the 5 point aOAMAT scale, with higher scores indicating more positive attitudes about MOUD ($SD = 0.66$) and 99.65 out of 144 on the ATP scale ($SD = 24.15$), with higher scores indicating more stigmatizing attitudes. With regard to beliefs about addiction, participants reported generally thinking that people with substance use problems cannot control their behavior ($M = 3.52$ out of 5, $SD = 0.76$), are responsible for their actions ($M = 4.23$ out of 5, $SD = 0.67$) and for recovery ($M = 4.33$ out of 5, $SD = 0.58$), that substances are used to cope ($M = 3.76$ out of 5, $SD = 0.57$), and that experts and professionals are needed in order to manage addiction ($M = 3.21$ out of 5, $SD = 0.86$). Participants reported limited beliefs that there is a genetic/biological basis for addiction ($M = 2.61$ out of 5, $SD = 0.82$)

Table 1
Descriptive statistics.

	N (%)	M (SD)	Actual Range
Variable			
Age		37.46 (8.15)	22–55
Race			0 – 1
White	123 (80.9%)		
Minority	29 (19.1%)		
Contact			1 – 3
Couple times/week	9 (5.9%)		
Every other day	3 (2.0%)		
Every day	140 (92.1%)		
Education			1 – 6
High School diploma	6 (3.9%)		
Vocational Degree	4 (2.6%)		
Some College	17 (11.2%)		
Bachelor's Degree	37 (24.3%)		
Master's Degree	35 (23.0%)		
Doctoral Degree	53 (34.9%)		
Agency			1 – 2
State/Federal Prison	128 (84.2%)		
Other Agency	24 (15.8%)		
Position			1 – 9
Correctional Officer	20 (13.2%)		
Counselor/Therapist	26 (17.1%)		
Psychologist	55 (36.2%)		
Psychiatrist	0 (0%)		
Social Worker/Case Manager	10 (6.6%)		
Health Care	10 (6.6%)		
Probation Officer	5 (3.3%)		
Administration Staff	21 (13.8%)		
Educator	3 (2.0%)		
Other	2 (1.3%)		
Years worked in Agency		103.99 (87.07)	6 – 372
Years worked with justice-involved individuals		132.70 (87.21)	9 – 372
Attitudes Toward Prisoners		99.65 (24.15)	36 – 144
ABI-cont (inability to control)		3.52 (0.76)	1 – 5
ABI-exp (reliance on experts)		3.21 (0.86)	1 – 5
ABI-act (responsibility for actions)		4.23 (0.67)	1 – 5
ABI-rec (responsibility for recovery)		4.33 (0.58)	1 – 5
ABI-gen (genetic basis)		2.61 (0.82)	1 – 5
ABI-cope (coping)		3.76 (0.57)	1 – 5
ABI-moral (moral weakness)		2.95 (0.75)	1 – 5
aOAMAT		3.41 (0.66)	1 – 5

Note: aOAMAT = adapted Opinions About Mediation Assisted Treatment scale

Table 2
Bivariate correlations.

	1	2	3	4	5	6	7	8	9	10	11	12
1. Age	–											
2. Race	-.05	–										
3. Education	-.19*	.12	–									
4. ATP	.10	-.01	-.47***	–								
5. ABI-cont	.35***	-.20*	-.03	-.06	–							
6. ABI-exp	.03	-.01	.05	.02	.35***	–						
7. ABI-act	.07	-.07	.13	.09	.30***	.09	–					
8. ABI-rec	-.09	-.04	.08	.22**	.20*	.23**	.37***	–				
9. ABI-gen	-.01	.09	.10	-.17*	-.04	-.001	-.34***	-.15	–			
10. ABI-cope	-.11	.11	.07	-.16*	.12	.22**	-.02	.10	.26**	–		
11. ABI-moral	.13	.06	-.35***	.69***	-.06	-.01	.19*	.38***	-.20*	-.02	–	
12. aOAMAT	-.13	-.01	.29**	-.46***	-.09	-.04	-.19*	-.16†	.23**	.09	-.41***	–

Note. * $p < .05$, ** $p < .01$, *** $p < .001$, † marginally significant. ATP = Attitudes Toward Prisoners; ABI-cont = Inability to control; ABI-exp = Reliance on experts; ABI-act = Responsibility for actions; ABI-rec = Responsibility for recovery; ABI-gen = Genetic basis; ABI-cope = Coping; ABI-moral = Moral weakness; aOAMAT = adapted Opinions About Mediation Assisted Treatment scale

and that addiction represents a moral weakness ($M = 2.95$ out of 5, $SD = 0.75$).

3.1. Correlations

Bivariate correlations are displayed in Table 2. Criminal legal staff with more education had more positive attitudes about MOUD, but no

other sociodemographic characteristics were significantly related to the aOAMAT. Believing that people with substance use problems are responsible for their actions and that addiction is a moral weakness were related to less positive attitudes about MOUD, and believing that there is a genetic/biological basis for addiction was related to more positive attitudes about MOUD. More stigmatizing attitudes about justice-involved people were related to more negative attitudes about MOUD.

Table 3

Regressions examining predictors of attitudes toward medications for opioid use disorder ($n = 141$).

Predictors	B (SE)	95% CI of B	t
Education	.05 (.04)	[-.03; .14]	1.24
Attitudes Toward Prisoners	-.01 (.003)*	[-.01; -.002]	-2.61
ABI-act (responsibility for actions)	-.05 (.09)	[-.23; .13]	-.53
ABI-rec (responsibility for recovery)	-.02 (.10)	[-.22; .17]	-.23
ABI-gen (genetic basis)	.08 (.07)	[-.05; .21]	1.25
ABI-moral (moral weakness)	-.13 (.10)	[-.32; .06]	-1.39

Note. * $p < .05$, ** $p < .01$, *** $p < .001$. Dependent variable: aOAMAT = adapted Opinions About Medication Assisted Treatment

3.2. Linear regression

Based on the presence of significant bivariate relationships with the aOAMAT, education, ABI-responsibility for actions, ABI-responsibility for recovery, ABI-moral weakness, ABI-genetic basis, and the ATP total score were simultaneously entered as predictors of the aOAMAT composite score in a multiple linear regression (see Table 3). Because education was highly related to position type, only years of education was included in analyses. Results showed that this model fit the data well ($F(134) = 7.48$, $p < .001$) and explained 25% of the variance in the aOAMAT. Among all predictors, only the ATP was significantly related to the aOAMAT ($\beta = -.27$, $p = .010$).

4. Discussion

The purpose of this study was to better understand what drives negative attitudes about MOUD among staff in the criminal legal system, who are often skeptical of the benefits of MOUD and unlikely to offer or recommend this treatment to justice-involved people. We hypothesized that among criminal legal staff, stigma toward criminal involvement and certain beliefs about addiction (e.g., it is a choice, moral weakness) would be associated with more negative attitudes toward MOUD. Results partially supported our hypotheses.

At the bivariate level, staff who believed individuals with addiction were responsible for their actions and recovery, morally weak, and did not have a genetic predisposition to substance use had more negative attitudes toward MOUD. Additionally, staff who had lower educational attainment had more negative attitudes toward MOUD. These findings are consistent with prior research which has shown that more education is typically associated with more positive MOUD attitudes (Andraka-Christou et al., 2019), and beliefs that addiction is a choice more than a disease are related to more negative MOUD attitudes (Richards et al., 2020). In a multiple regression model including demographics, addiction beliefs, and stigmatizing attitudes toward justice-involved people, only the latter was significantly associated with attitudes about MOUD. Thus, although responsibility for one's actions and recovery, beliefs around moral weakness, and genetic predisposition were relevant at the bivariate level, stigma toward "criminals" better explained negative attitudes about MOUD among criminal legal staff.

Stigma associated with criminal involvement is pervasive in the U.S., and it is not uncommon for justice-involved people to anticipate and/or experience judgement and rejection from family, friends, healthcare providers, criminal legal staff, and community members broadly (Moore & Tangey, 2017; LeBel et al., 2012; van Olphen et al., 2009). Much research has shown that criminal legal staff have stigmatizing attitudes about criminal involvement, sometimes more severe than the general public (Kjelsburg et al., 2007; Rade et al., 2016), and that this is associated with a preference for more punitive and less rehabilitative approaches in general (Forman et al., 2001; Schaefer & Williams, 2018). For instance, studies with criminal legal staff have shown that staff often agree that justice-involved people are untrustworthy, unmotivated, and unlikely to change (Craig, 2004; Kjelsburg et al., 2007). These types

of attitudes contribute to a negative environment that is not conducive to offering evidence-based treatments for substance use (Cooper & Neilson, 2017), and impact how staff interact with clients (Forman et al., 2001; Rudes et al., 2011).

4.1. Clinical implications

There have been increasing efforts to offer MOUD along the criminal legal system continuum to address overdose risk and reduce other negative health outcomes from OUD, including during incarceration and at release (Belenko et al., 2013; Brinkley-Rubinstein et al., 2018). As part of this movement, there has been a recent push to decrease stigma about OUD and MOUD among staff in all major systems, including the criminal legal system (Grella et al., 2020). However, a review of opioidlibrary.org resources for reducing stigma about MOUD shows that intervention approaches rarely address the criminal legal system, and the ones that do are not focused on stigma around criminal involvement. Only one intervention to our knowledge is being developed for criminal legal staff that addresses stigma associated with substance use, MOUD, and criminal involvement (authors, under review). Moreover, interventions to reduce stigmatizing attitudes in general suggest that psychoeducation as well as contact with members of the stigmatized group who are currently functioning adaptively can address negative attitudes (Corrigan et al., 2012). In fact, a recent review showed that including people in recovery in stigma reduction interventions is more effective than purely using a knowledge-based approach to reduce stigma (Bielenberg et al., 2021). To target the belief that incarcerated people cannot change, it may be similarly helpful to include people who are experiencing success post-incarceration, including those with lived experience of substance use disorder, in anti-stigma trainings geared toward criminal legal staff. Overall, the stigmatizing attitudes that criminal legal staff hold about criminal involvement have to be addressed in order to increase uptake of treatment approaches for addiction in this system.

4.2. Limitations and future directions

Although this study is the first to examine novel stigma drivers of MOUD attitudes among criminal legal staff, it is not without limitations. First, this study recruited a convenience sample that was predominantly white, mental health staff, who worked in state or federal prison settings. As of the latest Bureau of Justice Statistics' Census of State and Federal Correctional Facilities, approximately 66% of staff in these settings were correctional officers, with only about 10% being professional staff like mental health providers (Stephan, 2008). Therefore, these results may not be generalizable to all criminal legal staff in all types of positions or agencies. Secondly, the current study only assessed attitudes about MOUD in general and did not assess attitudes about each type of MOUD. To reduce assessment burden, methadone and buprenorphine were asked about together in each survey item which prevented us from being able to evaluate how these medications were viewed differently. However, there are important distinctions between these medications as well as naltrexone, which we did not assess. Criminal legal staff may have more negative attitudes about methadone and buprenorphine compared to naltrexone because of the potential to divert or misuse these medications (Andraka-Christou et al., 2019). Methadone is often considered the most stigmatized due to its euphoric effect, which can mimic a "high" similar to using opioids (Woo et al., 2017). Among U.S. drug courts that allow MOUD, methadone is most restricted and long-acting injectable naltrexone is thought to be the most accepted approach (Matusow et al., 2013). Future research should examine attitudes about each type of MOUD (i.e., separately asking about methadone, buprenorphine, and naltrexone), as well as consider additional predictors of MOUD attitudes, like experience treating people with OUD and perception of diversion. Furthermore, these data were collected via a moderate-length online survey (45–60 min) during the COVID-19 pandemic when many legal systems were strained and staff may have ex-

perienced assessment fatigue, which may have influenced responses. Lastly, we only assessed *attitudes*, and although criminal legal staffs' attitudes have been predictors of their behavior (Hogarth, 1974), it will be important to evaluate behaviors such as referral to MOUD, and implementation of MOUD as outcomes.

5. Conclusions

Stigma toward people with substance use disorders, those involved with the criminal legal system, and MOUD are pervasive and for many, "may literally be a matter of life or death" (Wakeman & Rich, 2018, pg. 332). There is a dearth of research examining drivers of MOUD attitudes among criminal legal staff. The current study helps address this gap and informs dissemination and implementation of MOUD in the criminal legal system. Specifically targeting the stigma of criminal involvement may facilitate uptake of MOUD with criminal legal staff.

Funding source

East Tennessee State University Research Development Committee. The funders had no role in study design; in the collection, analysis and interpretation of data; in the writing of the report; or in the decision to submit the article for publication. The views and opinions expressed are those of the authors.

Author disclosures

Role of Funding Source: Research was supported by the East Tennessee State University Research Development Committee; the funder had no role in the design, conduct, analysis, or manuscript development.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Kelly E. Moore: Conceptualization, Writing – original draft, Writing – review & editing. **Shania L. Siebert:** Formal analysis, Writing – original draft, Writing – review & editing. **Rachelle Kromash:** Writing – original draft, Writing – review & editing. **Mandy D. Owens:** Conceptualization, Writing – original draft, Writing – review & editing. **Diamond C. Allen:** Writing – original draft, Writing – review & editing.

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